

Guy Davidson

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I am a cognitive scientist excited about understanding the human mind and leveraging ideas from human cognition to develop more human-like artificial intelligence. My research combines human experiments, data analysis, and computational modeling to study cognition. I am particularly interested in cognitive goals and the use of structured, program-like representations to capture them. I have also led work on cognitively-inspired machine learning and contributed to more theoretical psychology efforts.

EDUCATION

NYU CENTER FOR DATA SCIENCE Ph.D. candidate

New York, NY
2019–Present

Advised by Professors Brenden Lake and Todd Gureckis, my thesis research attempts to advance the understanding of goals in human cognition. My work proposes a framework to represent goals as reward-producing programs and supports this idea by combining evidence from rich human experiments and computational models. My research offers a novel treatment of goals in cognitive science and aims to contribute to the development of richer goal representations for artificial agents.

BRAINS, MINDS, AND MACHINES SUMMER COURSE Hosted by the MIT and Harvard Center for Brains, Minds, and Machines

Woods Hole, MA
08/2021

MACHINE LEARNING SUMMER SCHOOL Hosted by University College London and Imperial College London

London, England
07/2019

MINERVA UNIVERSITY BSc in Computational Sciences

San Francisco, CA
2015–2019

I graduated summa cum laude with a concentration in Machine Learning **GPA: 3.98/4.0**. In my **capstone project**, I investigated the scaling behavior of different meta-learning algorithms: how quickly new tasks are learned as a function of previous training, using a novel benchmark paradigm inspired by visual question answering (externally advised by Mike Mozer, published at CVPR 2020).

PUBLICATIONS AND PRESENTATIONS

Davidson, G., Todd, G., Togelius, J., Gureckis, T. M., Lake, B. M. (2024). Goals as Reward-Producing Programs. *Soon to be submitted to Nature Machine Intelligence*: https://guydavidson.me/files/goals_as_programs.pdf

Davidson, G., Orhan, A. E., Lake, B. M. (2024). Spatial Relation Categorization in Infants and Deep Neural Networks. *Cognition*.

Sharma, S. **Davidson, G.**, Khetarpal, K., Kanervisto, A., Arora, U., Hofmann, K., Momennejad, I. (2024). Toward Human-AI Alignment in Large-Scale Multi-Player Games. *ArXiv*.

Davidson, G., Todd, G., Gureckis, T. M., Togelius, J., Lake, B. M. (2023). Generating Human-Like Goals by Synthesizing Reward-Producing Programs. *Intrinsically Motivated Open-ended Learning Workshop @ NeurIPS 2023*.

Davidson, G., Gureckis, T. M., & Lake, B. M. (2022). Creativity, Compositionality, and Common Sense in Human Goal Generation. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society, CogSci 2022*.

Davidson, G., Lake, B. M. (2021). Examining Infant Relation Categorization Through Deep Neural Networks. *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society, CogSci 2021*.

Bennett, D., **Davidson, G.**, & Niv, Y. (2021). A model of mood as integrated advantage. *Psychological Review*.

Davidson, G., Lake, B. M. (2020). Investigating Simple Object Representations in Model-Free Deep Reinforcement Learning. *Proceedings of the 42nd Annual Meeting of the Cognitive Science Society, CogSci 2020*.

Davidson, G., Lake, B. M. (2020). Systematically Comparing Neural Network Architectures in Relation Learning. *Object-Oriented Learning (OOL): Perception, Representation, and Reasoning Workshop @ ICML 2020*.

Davidson, G., Mozer, M. C. (2020). Sequential mastery of multiple visual tasks: Networks naturally learn to learn and forget to forget. *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Davidson, G., Radulescu, A., & Niv, Y. (2019). Contrasting the effects of prospective attention and retrospective decay in representation learning. *The 4th Multidisciplinary Conference on Reinforcement Learning and Decision Making*.

Bennett, D., **Davidson, G.**, & Niv, Y. (2019). Momentum and mood in policy-gradient reinforcement learning. *The 4th Multidisciplinary Conference on Reinforcement Learning and Decision Making*.

SELECTED INVITED TALKS

CoLab, Princeton University	05/2024
Invited talk, Cognitive Control of Action Workshop, Princeton University	03/2024
CoCoDev Lab, Harvard University	02/2024
CoCoSci Lab, Massachusetts Institute of Technology	02/2024

TEACHING

DS-GA 1016: COMPUTATIONAL COGNITIVE SCIENCE	NYU
Section leader, grader	01/2021-05/2021, 01/2022-05/2022
Served as a section leader and grader for graduate-level course focused on computational approaches to modeling cognition.	
DS-UA 112: INTRODUCTION TO DATA SCIENCE	NYU
Section leader	09/2019-12/2019
Served as a section leader for new undergraduate course introducing students to NYU's newly approved Data Science major.	

RESEARCH & PROFESSIONAL EXPERIENCE

MICROSOFT RESEARCH	New York, NY
Research Intern	05/2022-08/2022
Developed methods inspired by the cognitive psychology concept of task-sets (abstract task representations) to analyze and predict behavior in a large-scale gameplay dataset in a multiplayer game. Initial results highlighted consistent differences between players by their propensity to flee or attack in fight-or-flight scenarios. Mentored by Ida Momennejad and Harm van Seijen.	
PRINCETON NEUROSCIENCE INSTITUTE	Princeton, NJ
Research Intern	05/2018-08/2018
Interned with Professor Yael Niv's lab, to investigate human reinforcement learning (RL) in multidimensional environments:	
- Modeled data from previous experiments, making discoveries regarding the dissociable roles of attention and decay in human RL, and the efficacy of eye-tracking and fMRI-based attention measures. Project presented at RLDM 2019. - Implemented a reinforcement learning experiment in a customizable web platform, enabling data collection using Amazon Mechanical Turk and building a framework used by several current lab members to develop new experiments. - Developed a simulation environment for bandit problems to motivate work framing mood as a momentum variable.	
AIDOC MEDICAL	Tel Aviv, Israel
Research Engineer	05/2017-11/2017
Implemented research-supporting tools in Azure cloud environment to facilitate and expedite deep learning experimentation, reducing idea-to-experiment turnaround 10x from 1-2 hours to 5-10 minutes.	
AMAZON PRIME AIR	Seattle, WA
Software Development Engineer Intern	05/2016-08/2016
MINERVA PROJECT	San Francisco, CA
Software Development Engineer Intern	01/2016-05/2016
SIMILARWEB	Tel Aviv, Israel
Software Engineer	06/2015-09/2015
ISRAEL DEFENSE FORCES INTELLIGENCE BRANCH	Israel
Team leader, training instructor, software engineer	08/2008-10/2014
- Team leader: grew team of four software developers to ten, responsible for developing tools to enhance analysis capabilities and solve production issues. Reduced turnaround time more than 2x, contributed to Israel Defense Prize-winning project. - Instructor: managed two other instructors, training ten pupils in software engineering and computer networking. - Software engineer: built in-house data ingestion pipelines and analysis tools. Developed primarily in Python and Java and cultivated debugging and fault analysis expertise.	

SKILLS AND TECHNICAL COMPETENCIES

- Neural network research and development (mostly PyTorch)
- Data science and analysis (Numpy, Pandas, Sklearn, Matplotlib)
- Software engineering, architecture design, and testing
- Human participant research experiment design
- Web experiment development (React, vue, tailwind, Firebase)
- Research environment and stimulus design (Unity, Blender)